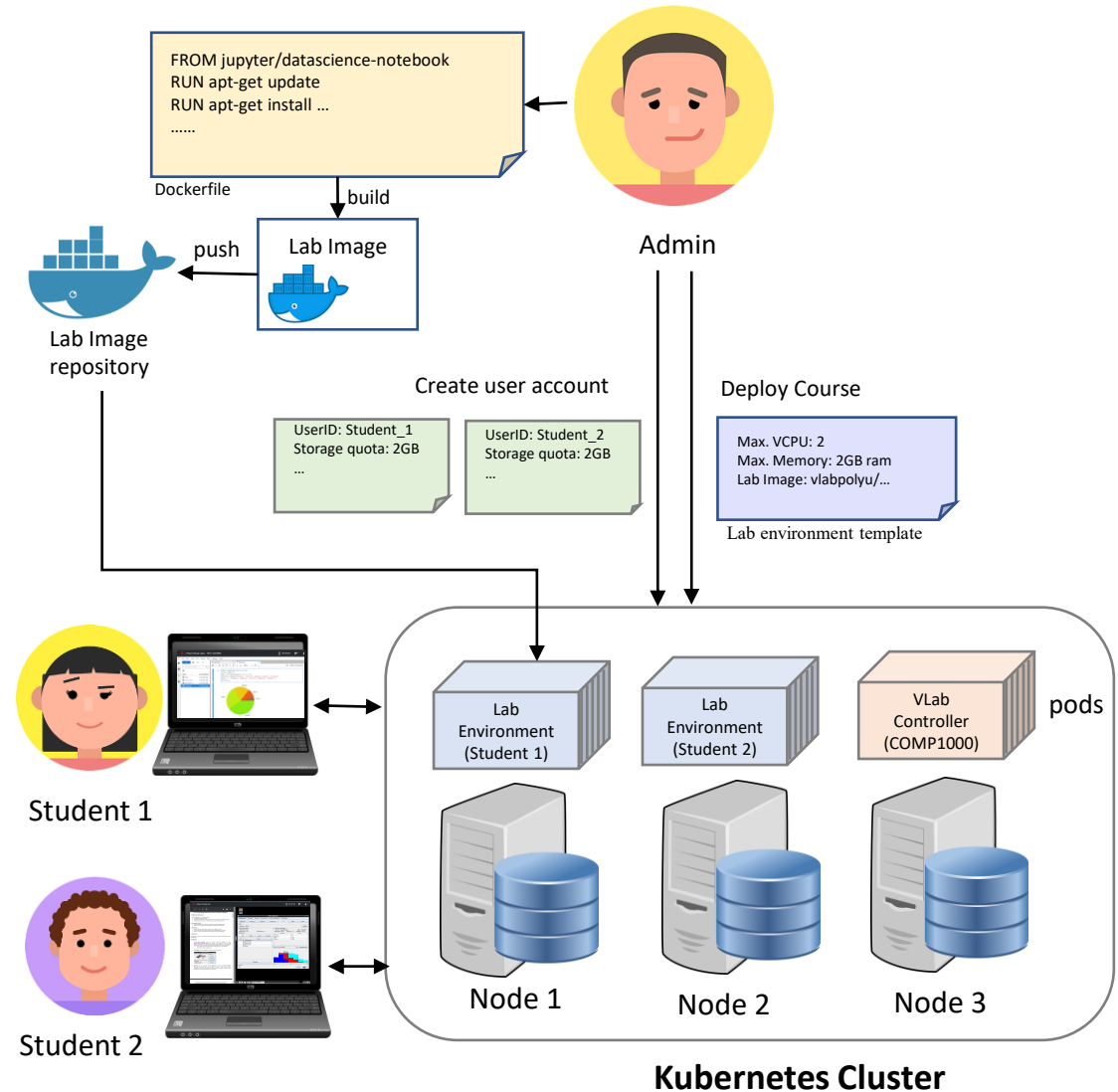


Development and adoption of a Cloud-based virtual lab platform

Richard Lui

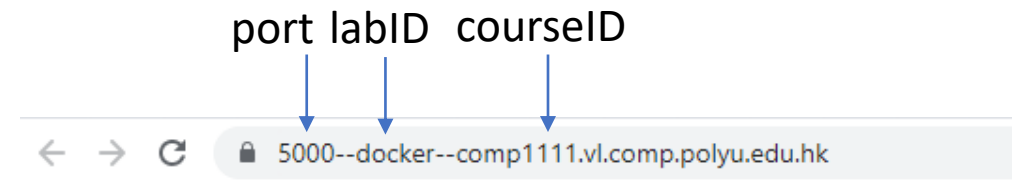
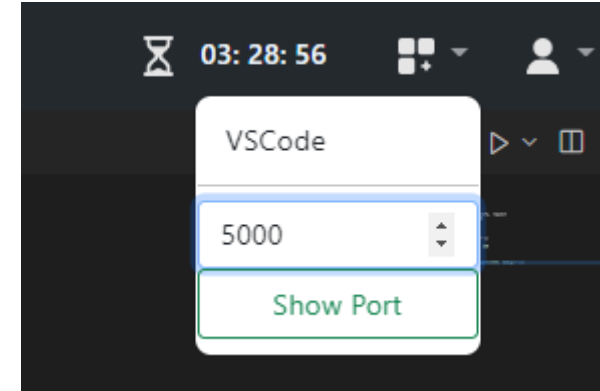
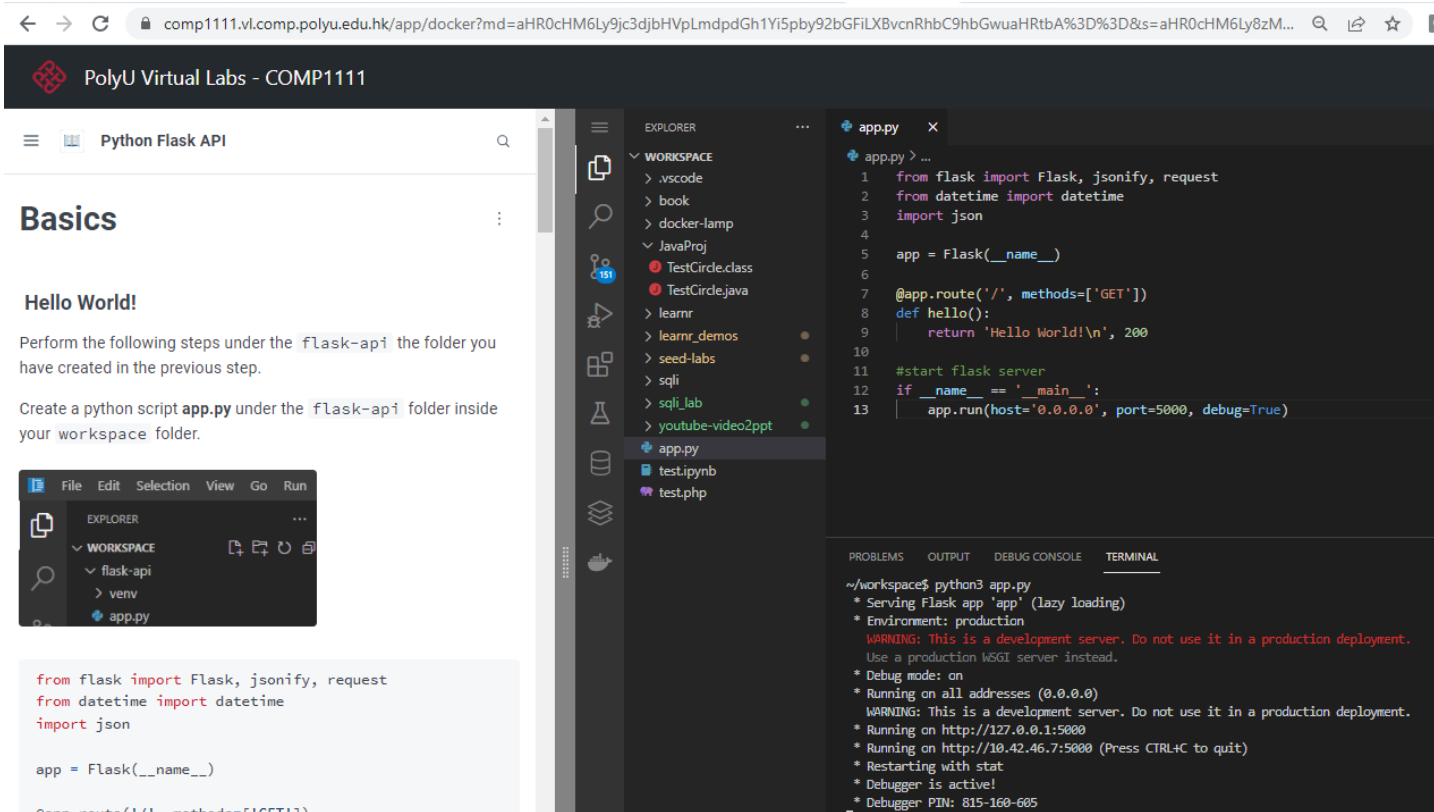
Background

- Challenges for conducting ‘hands-on’ labs related to computing
 - Students failed to properly set up the lab environment on their computers due to incompatibility with the operating system or other installed software
 - Struggle to complete the assigned tasks due to inconsistent settings
- Developed a prototype of a virtual lab platform (VLP) in our pilot project¹
 - Instructors can create customized lab environments for several computing-related courses
 - Students use a web browser to complete the lab tasks
 - Sandboxed lab and live coding environment deployed in the cloud.
 - No need to install custom software in the student’s local environment



1. “Practical class in VTL #3: Cloud-based coding laboratory for computing, AI, data analytics in Science and Engineering disciplines”

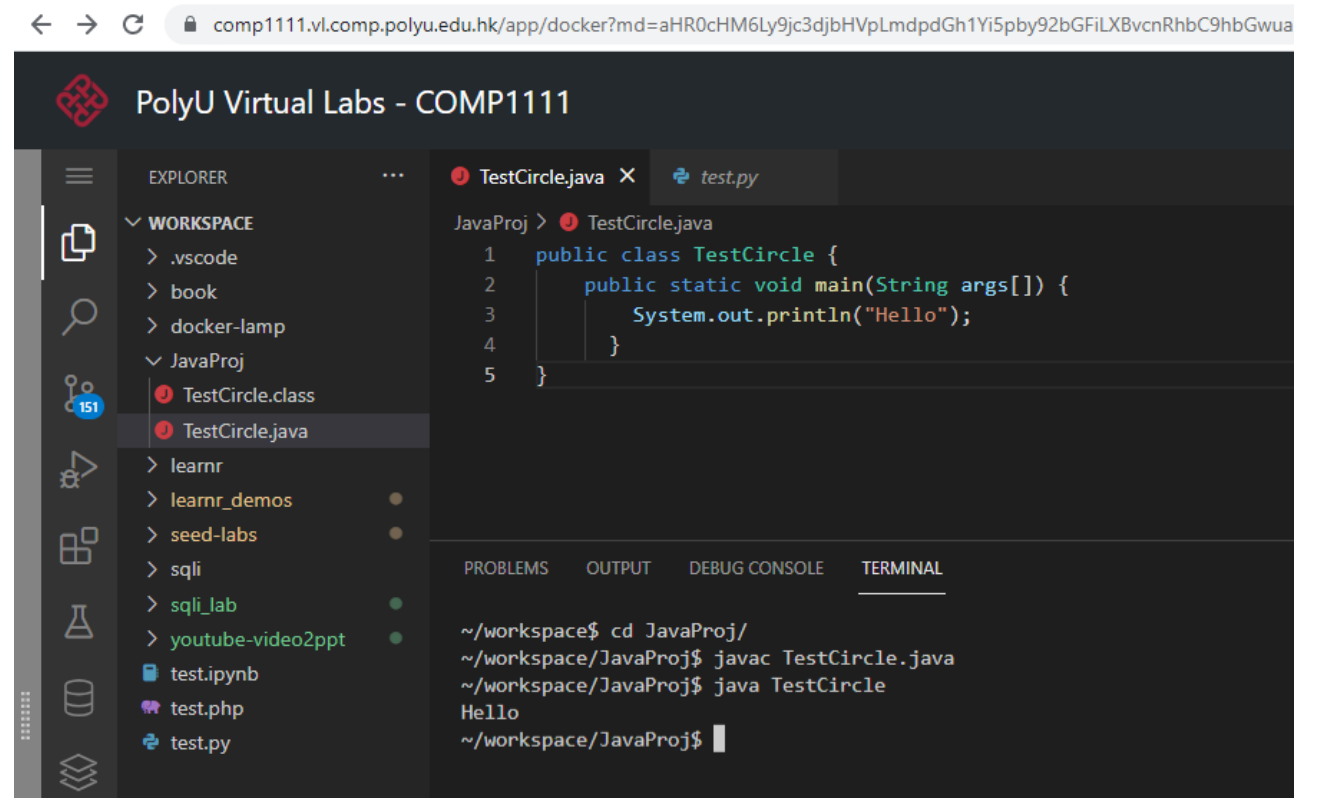
User Interface



Hello World!

Java Programming

- Java OOP, Maven, Java Spring, ...



comp1111.vl.comp.polyu.edu.hk/app/docker?md=aHR0cHM6Ly9jc3djbHVpLmdpdGh1Yi5pby92bGFpLXBvcnRhbnC9hbGwua

PolyU Virtual Labs - COMP1111

EXPLORER

WORKSPACE

- > .vscode
- > book
- > docker-lamp
- ▼ JavaProj
 - TestCircle.class
 - TestCircle.java
- > learnr
- > learnr_demos
- > seed-labs
- > sqli
- > sqli_lab
- > youtube-video2ppt
- test.ipynb
- test.php
- test.py

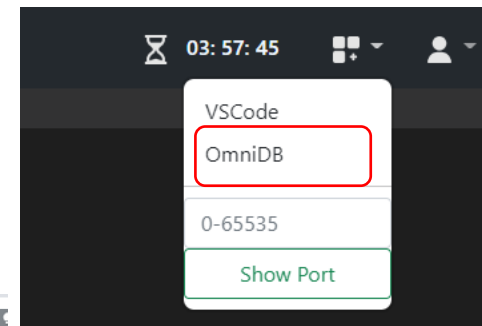
TestCircle.java

```
1 public class TestCircle {
2     public static void main(String args[]) {
3         System.out.println("Hello");
4     }
5 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
~/workspace$ cd JavaProj/
~/workspace/JavaProj$ javac TestCircle.java
~/workspace/JavaProj$ java TestCircle
Hello
~/workspace/JavaProj$
```

Databases



comp3335-2223-s1.vl.comp.polyu.edu.hk/app/oracle?md=aHR0cHM6Ly9wb2x5dS1jb21wLmdpdGJvb2suaW8vb3JhY2xIMThj&s=aHR0cHM6...

PolyU Virtual Labs - COMP3335-2223-S1

Oracle18c

Overview

Learning Objectives

- Basic commands for interacting with Oracle database
- Using SQLPlus
- Unlocking HR sample schema
- Creating pluggable databases
- Understanding Users and authorization in Oracle

Preparation

You should complete this exercise using the "Oracle 18c XE" image.

- The changes you have made in the database are not persistent across the different

```
File Edit Selection View Go Run Terminal Help workspace - code-server

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE

/root/workspace $ start-oracle
Starting Oracle Net Listener.
Oracle Net Listener started.
Starting Oracle Database instance XE.
Oracle Database instance XE started.

/root/workspace $ su oracle
/root/workspace $ sqlplus / as sysdba

SQL*Plus: Release 18.0.0.0.0 - Production on Wed Nov 23 14:23:42 2022
Version 18.4.0.0.0

Copyright (c) 1982, 2018, Oracle. All rights reserved.

Connected to:
Oracle Database 18c Express Edition Release 18.0.0.0.0 - Production
Version 18.4.0.0.0

SQL> select user from dual;

USER
-----
SYS
```

PolyU Virtual Labs - COMP3335-2223-S1

selected DB: XEPDB1

Oracle XE 18.0.0.0.0

XEPDB1

Tables (7)

- > COUNTRIES
- > DEPARTMENTS
- > EMPLOYEES
- > JOBS
- > JOB_HISTORY
- > LOCATIONS
- > REGIONS

Sequences

Views

Functions

Procedures

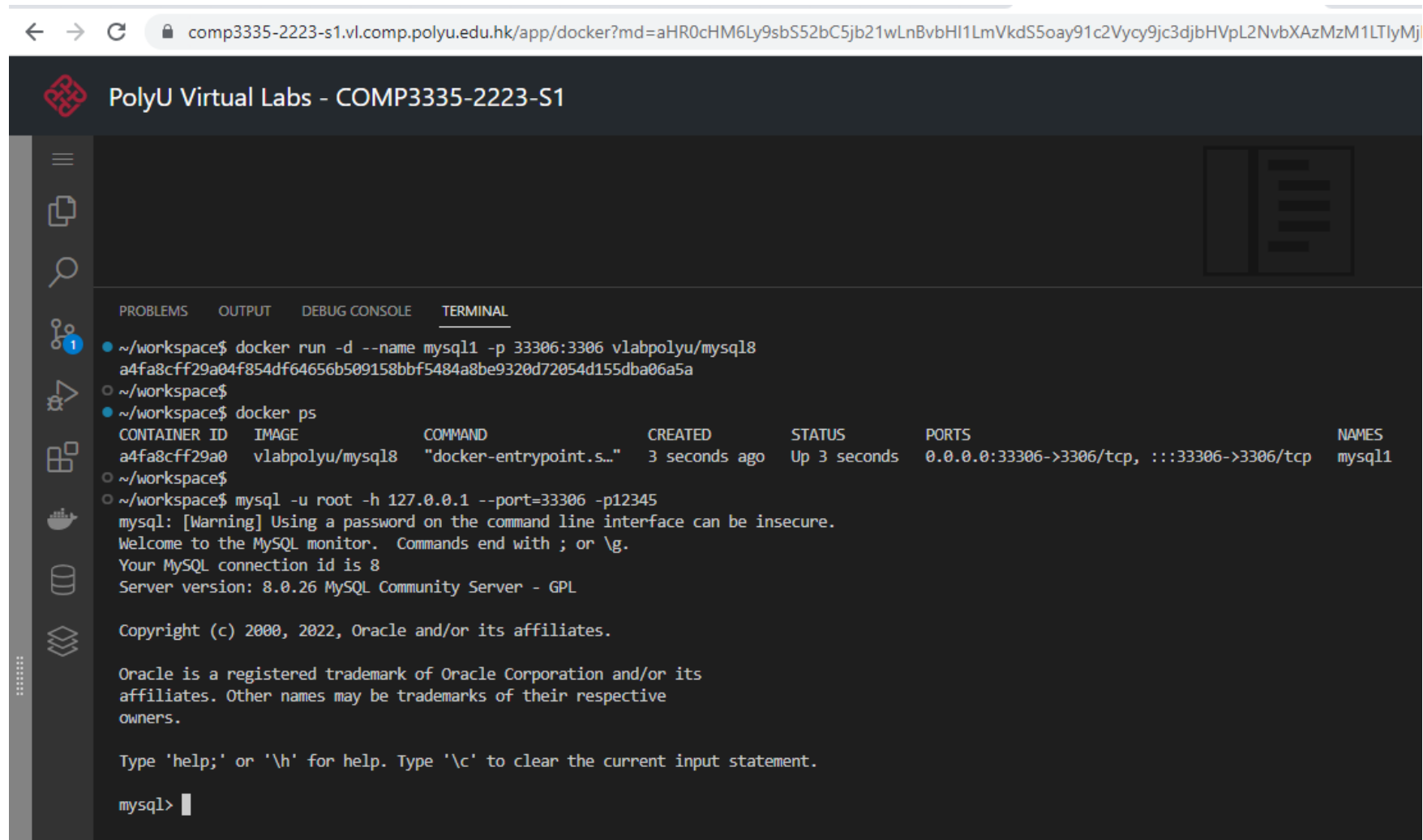
Console Query

```
1 select *
2 from employees;
```

50 rows in 17.665 ms
Start time: 11/04/2022 23:30:23

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMP
1	100	Steven	King	SKING	515.123.4567	2003-06-17 00:00:00	AD_PRES	24000	
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	2005-09-21 00:00:00	AD_VP	17000	
3	102	Lex	De Haan	LDEHAAN	515.123.4569	2001-01-13 00:00:00	AD_VP	17000	
4	103	Alexander	Hunold	AHUNOLD	590.423.4567	2006-01-03 00:00:00	IT_PROG	9000	
5	104	Bruce	Ernst	BERNST	590.423.4568	2007-05-21 00:00:00	IT_PROG	6000	
6	105	David	Austin	DAUSTIN	590.423.4569	2005-06-25 00:00:00	IT_PROG	4800	
7	106	Valli	Pataballa	VPATABAL	590.423.4560	2006-02-05 00:00:00	IT_PROG	4800	
8	107	Diana	Lorentz	DLORENTZ	590.423.5567	2007-02-07 00:00:00	IT_PROG	4200	
9	108	Nancy	Greenberg	NGREENBE	515.124.4569	2002-08-17 00:00:00	FL_MGR	12008	
10	109	Daniel	Faviet	DFAVIET	515.124.4169	2002-08-16 00:00:00	FL_ACCOUNT	9000	
11	110	John	Chen	JCHEN	515.124.4269	2005-09-28 00:00:00	FL_ACCOUNT	8200	
12	111	Ismael	Sciarra	ISCIARRA	515.124.4369	2005-09-30 00:00:00	FL_ACCOUNT	7700	

Running docker in the lab environment



```
comp3335-2223-s1.vl.comp.polyu.edu.hk/app/docker?md=aHR0cHM6Ly9sbS52bC5jb21wLnBvbHI1LmVkdS5oay91c2Vycy9jc3djbHVpL2NvbXAzMzM1LTlyMjI=

PolyU Virtual Labs - COMP3335-2223-S1

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

~/workspace$ docker run -d --name mysql1 -p 33306:3306 vlabpolyu/mysql8
a4fa8cff29a04f854df64656b509158bbf5484a8be9320d72054d155dba06a5a
~/workspace$
~/workspace$ docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED        STATUS        PORTS                               NAMES
a4fa8cff29a0   vlabpolyu/mysql8 "docker-entrypoint.s..." 3 seconds ago  Up 3 seconds  0.0.0.0:33306->3306/tcp, :::33306->3306/tcp  mysql1
~/workspace$
~/workspace$ mysql -u root -h 127.0.0.1 --port=33306 -p12345
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.26 MySQL Community Server - GPL

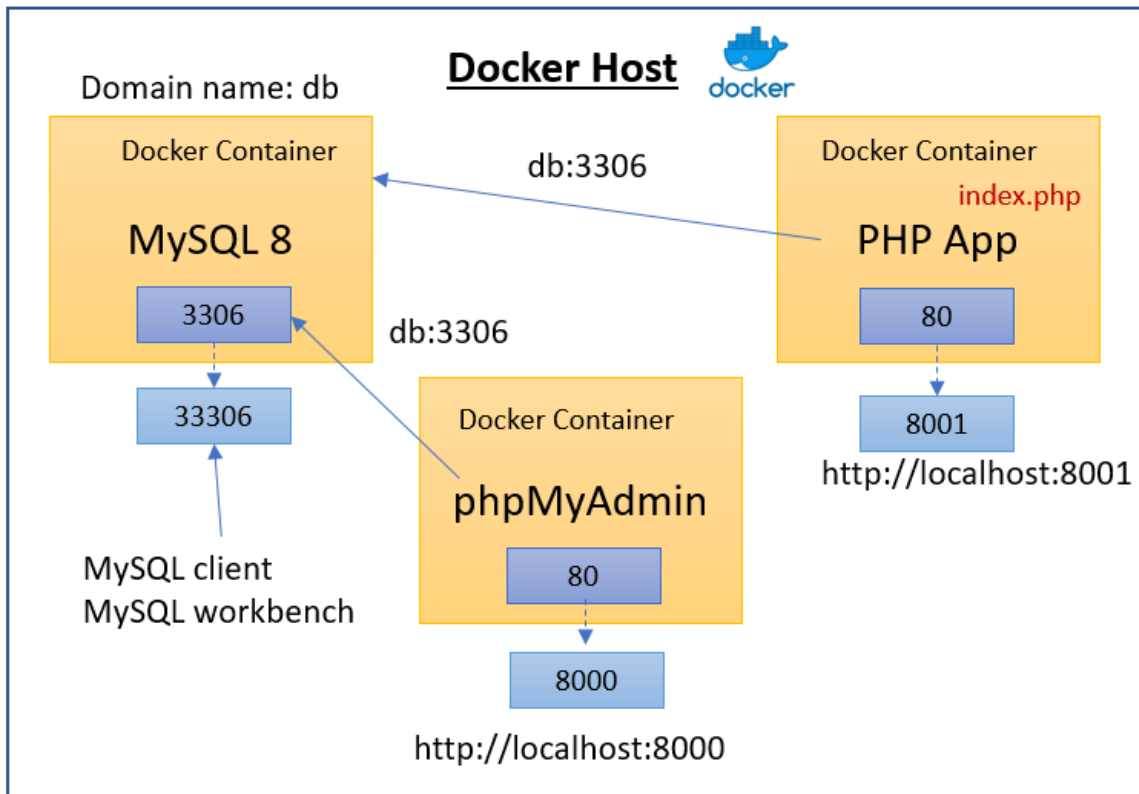
Copyright (c) 2000, 2022, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

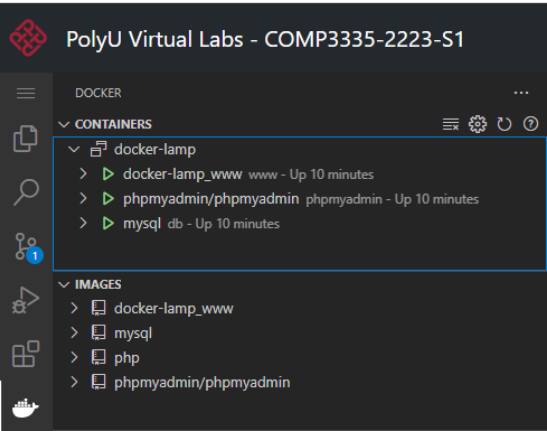
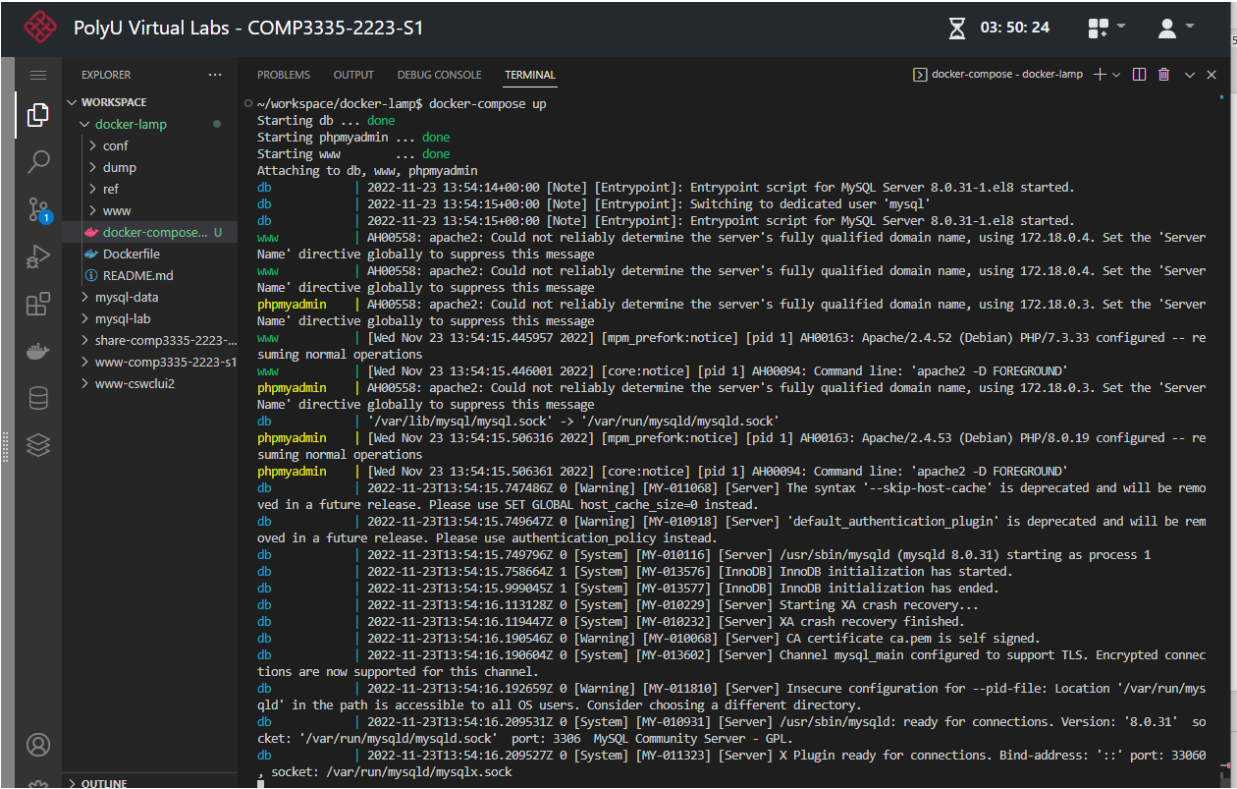
mysql>
```

Example: LAMP stack



```
docker-lamp > docker-compose.yml
1  version: "3.1"
2  services:
3
4      www:
5          build: .
6          container_name: www
7          ports:
8              - "8001:80"
9          volumes:
10             - ./www:/var/www/html/
11          depends_on:
12             - db
13
14      db:
15          image: mysql
16          container_name: db
17          command: --default-authentication-plugin=mysql_native_password
18          environment:
19              MYSQL_ROOT_PASSWORD: test
20              MYSQL_DATABASE: myDb
21              MYSQL_USER: user
22              MYSQL_PASSWORD: test
23          ports:
24              - "33306:3306"
25          volumes:
26              - ./dump:/docker-entrypoint-initdb.d
27              - ./conf:/etc/mysql/conf.d
28              - persistent:/var/lib/mysql
29
30      phpmyadmin: #Add this container
31          image: phpmyadmin/phpmyadmin
32          container_name: phpmyadmin
33          ports:
34              - 8000:80
35          environment:
36              PMA_HOST: 'db'
37              PMA_PORT: '3306'
38              PMA_USER: 'root'
39              PMA_PASSWORD: "test"
40          depends_on:
41              - db
```

• Launch a LAMP stack using docker-compose



port labID courseID

8001--docker--comp3335-2223-s1.vl.comp.polyu.edu.hk

Users

id	name	age
1	William	30
2	Marc	17
3	John	55

03: 48: 56

VSCode

8000

Show Port

8000--docker--comp3335-2223-s1.vl.comp.polyu.edu.hk

phpMyAdmin

Recent Favorites

New

- information_schema
- myDb
- mysql
- performance_schema
- sys

Server: db.3306

Databases SQL Status User accounts Export Import Settings Binary log More

General settings

Server connection collation: utf8mb4_unicode_ci

More settings

Appearance settings

Language: English

Theme: pmahomme View all

Database server

- Server: db via TCP/IP
- Server type: MySQL
- Server connection: SSL is not being used
- Server version: 8.0.31 - MySQL Community Server - GPL
- Protocol version: 10
- User: root@172.18.0.3
- Server charset: UTF-8 Unicode (utf8mb4)

Web server

- Apache/2.4.53 (Debian)
- Database client version: libmysql - mysqlnd 8.0.19
- PHP extension: mysql curl mbstring
- PHP version: 8.0.19

phpMyAdmin

- Version information: 5.2.0 (up to date)

Other lab environment

PolyU Virtual Labs - TEST-COURSE

File Edit View Run Kernel Tabs Settings Help

Filter files by name

Name	Last Modified
/	
data	2 months ago
www-cswd...	6 months ago
www-test-c...	4 minutes ago
test.ipynb	3 minutes ago

```
[2]: import matplotlib.pyplot as plt
x=[4,9,16,25,36]
fig = plt.figure(figsize =(5.5, 5.5))
plt.pie(x, labels= ("Guavas", "Berries", "Mangoes", "Apples", "Avocado"),
        colors = ( "#a8654d", "#eb5b13", "#ebc713", "#8deb13", "#8aeb13"))
plt.show()
```

The pie chart displays the distribution of five fruits. The slices are labeled as follows: Mangoes (yellow), Berries (orange), Guavas (brown), Apples (light green), and Avocado (dark green). The slices are arranged in a clockwise order starting from the top.

Fruit	Count
Guavas	4
Berries	9
Mangoes	16
Apples	25
Avocado	36

PolyU Virtual Lab

1 / 17 − + ↺ ↻ 🖨 ⋮

Lab 3: Data Mining using Weka on PolyU Virtual Lab Platform

A. Objectives and Outcomes

After finishing this lab, you should be able to

- Use Weka to perform data mining
- Understand the differences between classification, cluster analysis, and regression analysis.
- Understand the characteristics of different classifiers and clustering algorithms

B. Assessment Criteria

- Ability to produce correct outputs from classification and clustering algorithms.
- Ability to choose machine learning algorithms for different data mining tasks.

C. Submission

- Answer all the questions and include only the screen captures required.
- Copy and paste the screen capture of Weka to a Word file and convert it to PDF.
- Submit your report to Blackboard before the deadline specified in Blackboard.

D. Procedures

Note that Steps 2-19 are for Classification, that Steps 20-26 are for Clustering, and that Steps 27-30 are for Regression.

Classification

- Access online platform.** Access the PolyU Virtual Lab Platform via URL: <https://cs3112.cmp.polyu.edu.hk>. The username is your student ID. Use the "Forgot password" function to reset your password for the first-time login or if you forget your password later. Check your PolyU Email and follow the instructions to reset your password.
 - Click "Weka". (It may take some time for loading, please wait with patience.)
 - Double click "Weka" on the desktop.
- Start Explorer.** Click the "Explorer" button to launch Weka Explorer.

The screenshot shows the Weka Explorer window. The 'Preprocess' tab is selected. The 'Current relation' is 'Iris' with 150 instances and 5 attributes. The 'Selected attribute' is 'sepalwidth' with a histogram showing a distribution from 4.3 to 7.9. The 'Status' bar shows 'OK'.

1. Access online platform. Access the PolyU Virtual Lab Platform via URL: <https://cs3112.cmp.polyu.edu.hk>. The username is your student ID. Use the "Forgot password" function to reset your password for the first-time login or if you forget your password later. Check your PolyU Email and follow the instructions to reset your password.

- Click "Weka". (It may take some time for loading, please wait with patience.)
- Double click "Weka" on the desktop.

2. Start Explorer. Click the "Explorer" button to launch Weka Explorer.

The screenshot shows the 'Weka GUI Chooser' window. The 'Explorer' button is highlighted under the 'Applications' tab.

3. Input data. To open a data set, click the "Open file..." button and click on the "Home" directory at the top right corner, open "Desktop" -> "data" -> "weka" folder. Select the "Iris.arff" file to load the Iris dataset. The dataset contains 150 instances, divided into three classes. Each instance has 4 attributes plus a class label.

The screenshot shows the RStudio IDE interface. The main window displays a scatter plot of 'pressure' (y-axis, 0 to 800) versus 'time' (x-axis, 0 to 350). The plot shows a sharp increase in pressure starting around time 200. The R console shows the command 'plot(pressure)' and the message 'You can also embed plots, for example: plot(pressure, echo=FALSE)'. The environment pane on the right shows 'Environment is empty'. The file explorer on the bottom right shows the project files.

The screenshot displays the PolyU Virtual Labs (Test) - comp1234 interface, showing a workflow for data visualization. The workflow includes Data, Datasets, Data Table, and Scatter Plot. The Data Table window displays 150 instances of the Iris dataset with columns: sepal width, sepal length, petal length, and petal width. The Scatter Plot window shows a scatter plot of sepal length vs. sepal width, with data points colored by species: Iris-setosa (blue), Iris-versicolor (red), and Iris-virginica (green).

Architectural Design



High Available
Distributed Storage
(Ceph)



Network File System

Storages



Reverse Proxy
(Nginx)



VLP Router
(pfSense)



Load Balancer
(MetalLB)



Virtual Network
in K8s Cluster
(Calico)

Networking



Container Orchestration
Engine
(Kubernetes)



Kubernetes Lightweight
Distribution (K3s)



Container Runtime
(Containerd)



Data Visualization Platform
(Grafana)



Metric Aggregation & Alert
(Prometheus)



Log Aggregation
(Loki)

Monitoring



Single-Sign-On
(Keycloak)

Authentication



Image Registry
& Cache
(Harbor)



Image Sync
(Kube-fledged)

**Image
Management**



VLP Core
(VLabController)



VLP Dashboard

VLP Tools



MySQL



MongoDB

Databases



Virtual Lab Instances
(Generated by VLabController,
managed by VLP dashboard)

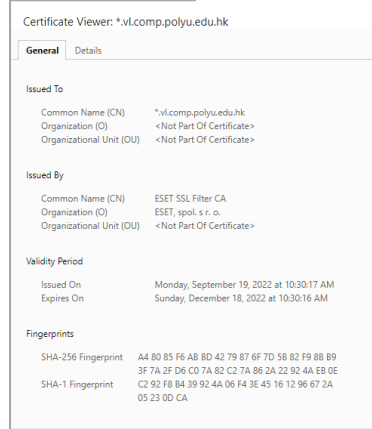
Network Architecture

Subdomain: *.vl.comp.polyu.edu.hk

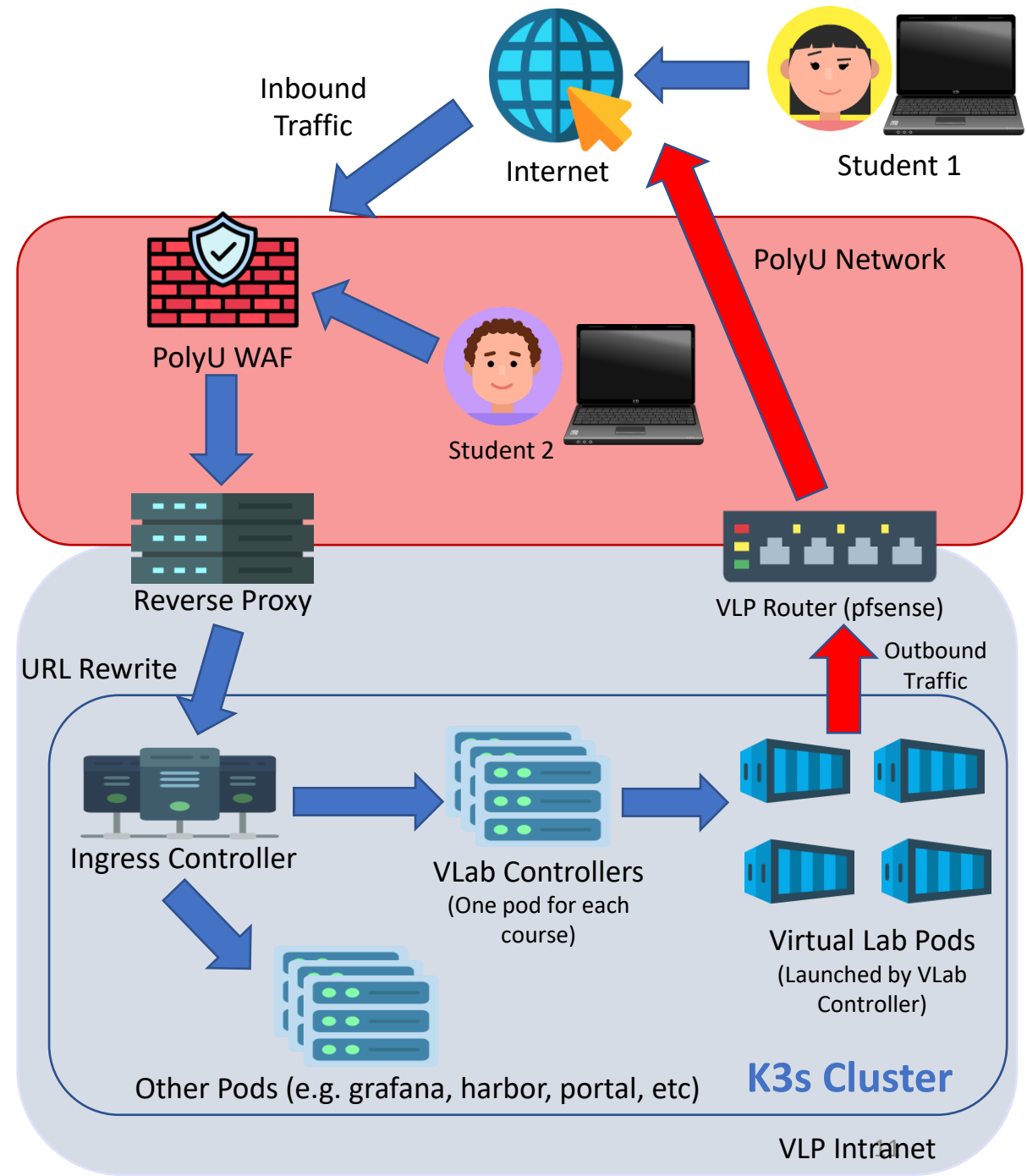


Hurricane Electric Free DNS Management Managing zone: vl.comp.polyu.edu.hk

Name	Type	TTL	Priority	Data
vl.comp.polyu.edu.hk	SOA	172800	-	ns1.he.net. hostmaster.he.net. 2021083100 86400 7200 3600000 172800
vl.comp.polyu.edu.hk	NS	300	-	ns1.he.net
vl.comp.polyu.edu.hk	NS	300	-	ns2.he.net
vl.comp.polyu.edu.hk	NS	300	-	ns3.he.net
vl.comp.polyu.edu.hk	NS	300	-	ns5.he.net
vl.comp.polyu.edu.hk	NS	300	-	ns4.he.net
localhost.vl.comp.polyu.edu.hk	A	300	-	127.0.0.1
my.vl.comp.polyu.edu.hk	A	300	-	158.132.159.73
*.vl.comp.polyu.edu.hk	CNAME	300	-	my.vl.comp.polyu.edu.hk



Host Overrides			
Host	Parent domain of host	IP to return for host	Description
master1	comp.polyu.edu.hk	192.168.1.3	kubernetes cluster master1
master2	comp.polyu.edu.hk	192.168.1.8	kubernetes cluster master2
nfs	comp.polyu.edu.hk	192.168.1.98	nfs server
reverse-proxy	comp.polyu.edu.hk	192.168.1.2	reverse proxy master
worker1	comp.polyu.edu.hk	192.168.1.4	kubernetes cluster worker1
worker2	comp.polyu.edu.hk	192.168.1.5	kubernetes cluster worker2
worker3	comp.polyu.edu.hk	192.168.1.6	kubernetes cluster worker3
worker4	comp.polyu.edu.hk	192.168.1.7	kubernetes cluster worker4



Kubernetes resources

Each student has 2GB quota of persistent storage (storing the user data in his/her home folder)

NAMESPACE1	NAME	STATUS	VOLUME	CAPACITY	ACCESS MODES	STORAGECLASS
vlab-22002446d	user-data	Bound	pvc-980d1fde8-1c3e-4215-95a3-3a22a2a7a0ca	2Gi	RWX	rook-cephfs
vlab-22086617d	user-data	Bound	pvc-a4b80cd1-3e2b-44e2-9ba8-7869a3889a0d	2Gi	RWX	rook-cephfs
vlab-22086694d	user-data	Bound	pvc-4ffccab1-a628-4fed-arc3f-bf18f87c0216	2Gi	RWX	rook-cephfs
vlab-22032832d	user-data	Bound	pvc-72dd06ed-d457-43d2-9f8d-38b677cbdc58	2Gi	RWX	rook-cephfs
vlab-22033037d	user-data	Bound	pvc-00bbbb43-d85f-45d2-b982-c8763e66549a	2Gi	RWX	rook-cephfs
vlab-22033907d	user-data	Bound	pvc-980fcea7-09f5-4182-9906-785eef094a87	2Gi	RWX	rook-cephfs
vlab-22033974d	user-data	Bound	pvc-f90f2c90-9584-4203-adef-2f51c8b184c1	2Gi	RWX	rook-cephfs
vlab-22063657d	user-data	Bound	pvc-8cd146d4-ef82-4bf5-95c8-ca890f60697c	2Gi	RWX	rook-cephfs
vlab-22058692d	user-data	Bound	pvc-d263b555-7735-44ac-bc74-e3bf3f3847b3	2Gi	RWX	rook-cephfs
vlab-22061954d	user-data	Bound	pvc-576dd0c2-1e0b-40eb-9f5b-469df6408f3c	2Gi	RWX	rook-cephfs
vlab-22068882d	user-data	Bound	pvc-89e8703f-9e06-42ad-a93c-24131c36912a	2Gi	RWX	rook-cephfs
vlab-22074136d	user-data	Bound	pvc-59f4f2f7-b091-4f43-b716-b8649e980444	2Gi	RWX	rook-cephfs
vlab-22074549d	user-data	Bound	pvc-bd729954-4461-476b-f68a7b9e4885	2Gi	RWX	rook-cephfs
vlab-22074541d	user-data	Bound	pvc-142521a6-cac2-4215-bf1f-c73c285cf47d	2Gi	RWX	rook-cephfs
vlab-22075379d	user-data	Bound	pvc-1a154b55-e9f4-473d-a358-bcf120aa2665	2Gi	RWX	rook-cephfs
vlab-22075432d	user-data	Bound	pvc-0dcdf0f8-b078-4228-a3d7-9e5cc077b59e	2Gi	RWX	rook-cephfs
vlab-22075554d	user-data	Bound	pvc-f0a42c4f-6a71-4d31-a9b6-55f3a140cd5	2Gi	RWX	rook-cephfs
vlab-22076083d	user-data	Bound	pvc-f3c21777-be91-4f15-a7cd-3b7d8bf851be	2Gi	RWX	rook-cephfs
vlab-22076247d	user-data	Bound	pvc-5c54583f-b9e4-435a-a7dd-99b209017e37	2Gi	RWX	rook-cephfs
vlab-22076316d	user-data	Bound	pvc-18c0a1a2-b882-4cfc-86b9-015829924d88	2Gi	RWX	rook-cephfs
vlab-22076673d	user-data	Bound	pvc-48c06e4-2c88-4d5e-9a29-442e8bd98f55	2Gi	RWX	rook-cephfs
vlab-22076919d	user-data	Bound	pvc-e3e3b131-8771-438b-ae9f-9e188ce99499	2Gi	RWX	rook-cephfs
vlab-22077019d	user-data	Bound	pvc-762079196-9d4c-4801-b919-7d51bf8802ed	2Gi	RWX	rook-cephfs
vlab-22077863d	user-data	Bound	pvc-a0a3b7cf-ae32-4a62-8975-c7fa9d8b6b6f	2Gi	RWX	rook-cephfs
vlab-22077115d	user-data	Bound	pvc-13a2f774-c8dc-4fc6-9f1f-086c3ff0d3d8	2Gi	RWX	rook-cephfs

Each of the student's lab environment is running as a kubernetes pod (with one or more containers).

The students' pod is mounted to his pvc.

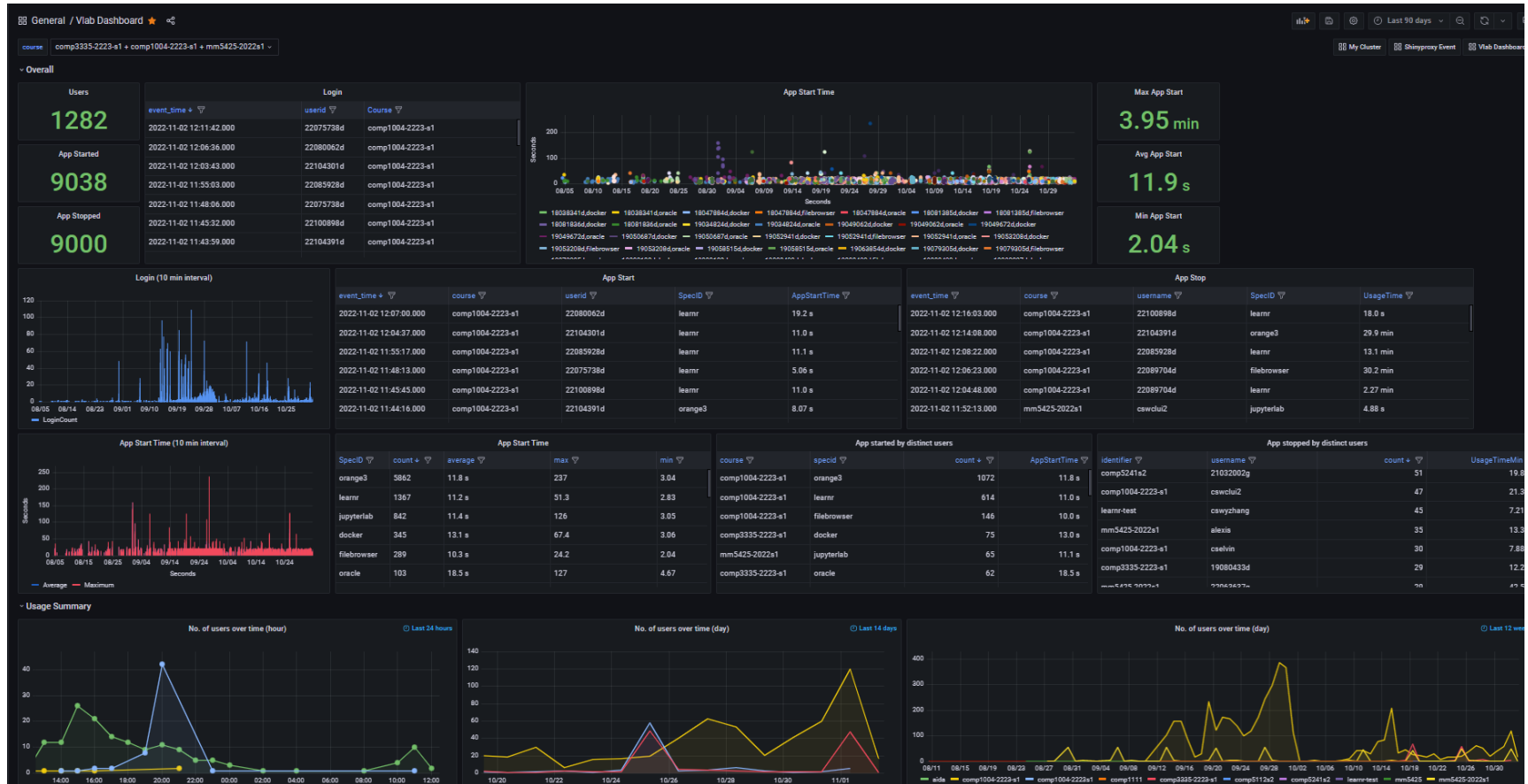
```
user@Dev: ~$ kubectl get pods --all-namespaces -o wide
Context: default
Cluster: default
User: default
K9s Rev: v0.24.15
K8s Rev: v1.21.3+k3s1
CPU: 11%
MEM: 12%

user@Dev: ~$ kubectl get pods --all-namespaces -o wide
Context: default
Cluster: default
User: default
K9s Rev: v0.24.15
K8s Rev: v1.21.3+k3s1
CPU: 11%
MEM: 12%

user@Dev: ~$ kubectl get pods --all-namespaces -o wide
Context: default
Cluster: default
User: default
K9s Rev: v0.24.15
K8s Rev: v1.21.3+k3s1
CPU: 11%
MEM: 12%

user@Dev: ~$ kubectl get pods --all-namespaces -o wide
Context: default
Cluster: default
User: default
K9s Rev: v0.24.15
K8s Rev: v1.21.3+k3s1
CPU: 11%
MEM: 12%
```

Platform Usage (Semester 1, 2022-2023)



Running Pod Resource utilization							
CourseID ▾	UserID ▾	SpecID ▾	Memory ▾	CPU ▾	Network transmitted ▾	Network received ▾	Uptime(min) ▾
comp3335-2223-s1	cswhu2	docker	1.22 GiB	0.0172	21.2 MiB	495 MiB	48.8
comp3335-2223-s1	cswhu2	oracle	2.25 GiB	0.0295	7.61 MiB	1.25 MiB	10.8

Example: Lab session in COMP3335

- 60+ students launching their own Oracle 18c database

